

JEE-2008
Paper II

1. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1 . From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i} + \hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in anticlockwise direction on a circle with centre at origin, to reach a point z_2 . The point z_2 is given by
- (A) $6 + 7i$ (B) $-7 + 6i$ (C) $7 + 6i$ (D) $-6 + 7i$

Answer

- ☐ (A) ☐ (B) ☐ (C) ☒ (D)

2. Let the function $g : (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is
- (A) even and is strictly increasing in $(0, \infty)$
 (B) odd and is strictly decreasing in $(-\infty, \infty)$
 (C) odd and is strictly increasing in $(-\infty, \infty)$
 (D) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

Answer

- ☐ (A) ☐ (B) ☒ (C) ☐ (D)

3. Consider a branch of the hyperbola

$$x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$$

with vertex at the point A . Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A , then the area of the triangle ABC is

- (A) $1 - \sqrt{\frac{2}{3}}$ (B) $\sqrt{\frac{3}{2}} - 1$ (C) $1 + \sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{3}{2}} + 1$

Answer

- ☐ (A) ☒ (B) ☐ (C) ☐ (D)

4. The area of the region between the curves $y = \sqrt{\frac{1 + \sin x}{\cos x}}$ and $y = \sqrt{\frac{1 - \sin x}{\cos x}}$ bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is

- (A) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$ (B) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$
 (C) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$ (D) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

Answer

- ☐ (A) ☒ (B) ☐ (C) ☐ (D)

5. Consider three points $P = (-\sin(\beta - \alpha), -\cos \beta)$, $Q = (\cos(\beta - \alpha), \sin \beta)$ and $R = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta))$, where $0 < \alpha, \beta, \theta < \frac{\pi}{4}$. Then,

- (A) P lies on the line segment RQ
 (B) Q lies on the line segment PR
 (C) R lies on the line segment QP
 (D) P, Q, R are non-collinear

Answer

- ☐ (A) ☐ (B) ☐ (C) ☒ (D)

6. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is

- (A) 2, 4 or 8 (B) 3, 6 or 9 (C) 4 or 8 (D) 5 or 10

Answer

- ☐ (A) ☐ (B) ☐ (C) ☒ (D)

7. Let two non-collinear unit vectors $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\vec{OP}$ (where O is the origin) is given by $\hat{a} \cos t + \hat{b} \sin t$. When P is farthest from origin O , let M be the length of $\vec{OP}$ and $\hat{u}$ be the unit vector along $\vec{OP}$. Then,

- (A) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$ (B) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$
 (C) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$ (D) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$

Answer

- ☒ (A) ☐ (B) ☐ (C) ☐ (D)

8. Let

$$I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx, \quad J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx.$$

Then, for an arbitrary constant C , the value of $J - I$ equals

- (A) $\frac{1}{2} \log \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right) + C$ (B) $\frac{1}{2} \log \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + C$
 (C) $\frac{1}{2} \log \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + C$ (D) $\frac{1}{2} \log \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + C$

Answer

- ☐ (A) ☐ (B) ☒ (C) ☐ (D)

9. Let $g(x) = \log f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = x f(x)$. Then, for $N = 1, 2, 3, \dots$,

$$g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$$

- (A) $-4\left\{1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2}\right\}$ (B) $4\left\{1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2}\right\}$
 (C) $-4\left\{1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2}\right\}$ (D) $4\left\{1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2}\right\}$

Answer

- ☒ (A) ☐ (B) ☐ (C) ☐ (D)

10. Suppose four distinct positive numbers a_1, a_2, a_3, a_4 are in G.P. Let $b_1 = a_1$, $b_2 = b_1 + a_2$, $b_3 = b_2 + a_3$ and $b_4 = b_3 + a_4$.

STATEMENT-1: The numbers b_1, b_2, b_3, b_4 are neither in A.P. nor in G.P.

and

STATEMENT-2: The numbers b_1, b_2, b_3, b_4 are in H.P.

- (A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1
 (B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
 (C) STATEMENT-1 is True, STATEMENT-2 is False
 (D) STATEMENT-1 is False, STATEMENT-2 is True

Answer

- ☐ (A) ☐ (B) ☒ (C) ☐ (D)

11. Let a, b, c, p, q be real numbers. Suppose α, β are the roots of the equation $x^2 + 2px + q = 0$ and $\alpha, \frac{1}{\beta}$ are the roots of the equation $ax^2 + 2bx + c = 0$, where $\beta^2 \notin \{-1, 0, 1\}$.

STATEMENT-1: $(p^2 - q)(b^2 - ac) \geq 0$

and

STATEMENT-2: $b \neq pa$ or $c \neq qa$

- (A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1
 (B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
 (C) STATEMENT-1 is True, STATEMENT-2 is False
 (D) STATEMENT-1 is False, STATEMENT-2 is True

Answer

- ☐ (A) ☒ (B) ☐ (C) ☐ (D)

12. Consider

$$L_1 : 2x + 3y + p - 3 = 0$$

$$L_2 : 2x + 3y + p + 3 = 0,$$

where p is a real number, and $C : x^2 + y^2 + 6x - 10y + 30 = 0$.

STATEMENT-1 : If line L_1 is a chord of circle C , then line L_2 is not always a diameter of circle C .

and

STATEMENT-2 : If line L_1 is a diameter of circle C , then line L_2 is not a chord of circle C .

- (A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is **NOT** a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

Answer



(A)

(B)

(C)

(D)

13. Let a solution $y = y(x)$ of the differential equation

$$x\sqrt{x^2 - 1} \, dy - y\sqrt{y^2 - 1} \, dx = 0$$

satisfy $y(2) = \frac{2}{\sqrt{3}}$.

$$\text{STATEMENT-1 : } y(x) = \sec\left(\sec^{-1} x - \frac{\pi}{6}\right)$$

and

STATEMENT-2 : $y(x)$ is given by

$$\frac{1}{y} = \frac{2\sqrt{3}}{x} - \sqrt{1 - \frac{1}{x^2}}$$

- (A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is **NOT** a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

Answer



(A)

(B)

(C)

(D)

Paragraph for Question Nos. 14 to 16

Consider the function $f : (-\infty, \infty) \rightarrow (-\infty, \infty)$ defined by

$$f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}, \quad 0 < a < 2.$$

14. Which of the following is true?

- (A) $(2+a)^2 f''(1) + (2-a)^2 f''(-1) = 0$ (B) $(2-a)^2 f''(1) - (2+a)^2 f''(-1) = 0$
(C) $f'(1)f'(-1) = (2-a)^2$ (D) $f'(1)f'(-1) = -(2+a)^2$

Answer

- ☒ ☐ ☐ ☐
(A) (B) (C) (D)

15. Which of the following is true?

- (A) $f(x)$ is decreasing on $(-1, 1)$ and has a local minimum at $x = 1$
(B) $f(x)$ is increasing on $(-1, 1)$ and has a local maximum at $x = 1$
(C) $f(x)$ is increasing on $(-1, 1)$ but has neither a local maximum nor a local minimum at $x = 1$
(D) $f(x)$ is decreasing on $(-1, 1)$ but has neither a local maximum nor a local minimum at $x = 1$

Answer

- ☒ ☐ ☐ ☐
(A) (B) (C) (D)

16. Let

$$g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt.$$

Which of the following is true?

- (A) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$
(B) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$
(C) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$
(D) $g'(x)$ does not change sign on $(-\infty, \infty)$

Answer

- ☐ ☒ ☐ ☐
(A) (B) (C) (D)

Paragraph for Question Nos. 17 to 19

Consider the lines

$$L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}$$

$$L_2 : \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$$

17. The unit vector perpendicular to both L_1 and L_2 is

- (A) $\frac{-\hat{i} + 7\hat{j} + 7\hat{k}}{\sqrt{99}}$ (B) $\frac{-\hat{i} - 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$
- (C) $\frac{-\hat{i} + 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$ (D) $\frac{7\hat{i} - 7\hat{j} - \hat{k}}{\sqrt{99}}$

Answer

- ☐ (A) ☒ (B) ☐ (C) ☐ (D)

18. The shortest distance between L_1 and L_2 is

- (A) 0 (B) $\frac{17}{\sqrt{3}}$ (C) $\frac{41}{5\sqrt{3}}$ (D) $\frac{17}{5\sqrt{3}}$

Answer

- ☐ (A) ☐ (B) ☐ (C) ☒ (D)

19. The distance of the point $(1, 1, 1)$ from the plane passing through the point $(-1, -2, -1)$ and whose normal is perpendicular to both the lines L_1 and L_2 is

- (A) $\frac{2}{\sqrt{75}}$ (B) $\frac{7}{\sqrt{75}}$ (C) $\frac{13}{\sqrt{75}}$ (D) $\frac{23}{\sqrt{75}}$

Answer

- ☐ (A) ☐ (B) ☒ (C) ☐ (D)

20. Consider the lines given by

$$L_1 : x + 3y - 5 = 0$$

$$L_2 : 3x - ky - 1 = 0$$

$$L_3 : 5x + 2y - 12 = 0$$

Match the Statements / Expressions in **Column I** with the Statements / Expressions in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I

Column II

- (A) L_1, L_2, L_3 are concurrent, if
- (B) One of L_1, L_2, L_3 is parallel to at least one of the other two, if
- (C) L_1, L_2, L_3 form a triangle, if
- (D) L_1, L_2, L_3 do not form a triangle, if

(p) $k = -9$

(q) $k = -\frac{6}{5}$

(r) $k = \frac{5}{6}$

(s) $k = 5$

Answer

	p	q	r	s
A	☐	☐	☐	☒
B	☒	☒	☐	☐
C	☐	☐	☒	☐
D	☒	☒	☐	☒

21. Match the Statements / Expressions in **Column I** with the Statements / Expressions in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I

Column II

- (A) The minimum value of $\frac{x^2 + 2x + 4}{x + 2}$ is
- (B) Let A and B be 3×3 matrices of real numbers, where A is symmetric, B is skew-symmetric, and $(A + B)(A - B) = (A - B)(A + B)$. If $(AB)^t = (-1)^k AB$, where $(AB)^t$ is the transpose of the matrix AB , then the possible values of k are
- (C) Let $a = \log_3 \log_3 2$. An integer k satisfying $1 < 2^{(-k + 3^{-a})} < 2$, must be less than
- (D) If $\sin \theta = \cos \varphi$, then the possible values of $\frac{1}{\pi}(\theta \pm \varphi - \frac{\pi}{2})$ are

(p) 0

(q) 1

(r) 2

(s) 3

Answer

	p	q	r	s
A	☐	☐	☒	☐
B	☐	☒	☐	☒
C	☐	☐	☒	☒
D	☒	☐	☒	☐

22. Consider all possible permutations of the letters of the word ENDEANOEL.

Match the Statements / Expressions in **Column I** with the Statements / Expressions in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I

- (A) The number of permutations containing the word ENDEA is
- (B) The number of permutations in which the letter E occurs in the first and the last positions is
- (C) The number of permutations in which none of the letters D, L, N occurs in the last five positions is
- (D) The number of permutations in which the letters A, E, O occur only in odd positions is

Column II

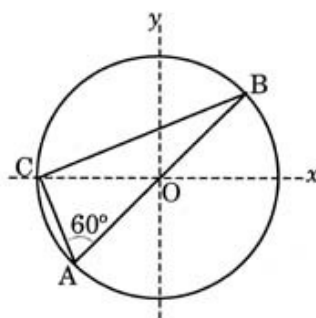
- (p) $5!$
- (q) $2 \times 5!$
- (r) $7 \times 5!$
- (s) $21 \times 5!$

Answer

	p	q	r	s
A	☒	☐	☐	☐
B	☐	☐	☐	☒
C	☐	☒	☐	☐
D	☐	☒	☐	☐

23. Consider a system of three charges $\frac{q}{3}$, $\frac{q}{3}$ and $-\frac{2q}{3}$ placed at points A, B and C, respectively, as shown in the figure. Take O to be the centre of the circle of radius R and angle $CAB = 60^\circ$

Figure :



- (A) The electric field at point O is $\frac{q}{8\pi\epsilon_0 R^2}$ directed along the negative x -axis
- (B) The potential energy of the system is zero
- (C) The magnitude of the force between the charges at C and B is $\frac{q^2}{54\pi\epsilon_0 R^2}$
- (D) The potential at point O is $\frac{q}{12\pi\epsilon_0 R}$

Answer

- ☐ ☐ ☒ ☐
- (A) (B) (C) (D)